INCH-POUND

MIL-S-29208B(YD)
17 January 1992
SUPERSEDING
MIL-S-29208A(YD)
27 March 1986

MILITARY SPECIFICATION

SPREADER, MATERIAL, POWER DRIVEN, TRUCK MOUNTED, COMMERCIAL

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hydraulically driven material spreader for spreading sand and aggregates or abrasive materials on road surfaces.

1.2 Classification. Spreaders shall be of the following types, as specified (see 6.2).

Type I - Truck Chassis Mounted.
Type II - Dump Body Mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

KKK-T-2108 - Truck and Truck Tractors: Commercial, Diesel and Gasoline Engine Driven, 24,000 to 32,000 GVW, 4 X 2.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The spreader shall consist of an all steel hopper type body complete with power driven auger or conveyor, rear-end material spreader and hydraulic system. The spreader shall be complete with all components and accessories to provide a complete and serviceable unit.

* 3.2 First article. When specified (see 6.2) in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.3 Standard commercial product. The spreader shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the spreader being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Safety. All rotating or reciprocating parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded.

* 3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.6 Construction. The equipment shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.7 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.8 Components.

3.8.1 Hopper body. The hopper body shall have a minimum water level capacity of 4.8 cubic yards. The inside length of the hopper shall be not less than 9 feet. The inside width of the hopper shall be not more than 80 inches. The hopper body shall be all-welded construction of not less than 10 gage steel. Body shall have a minimum of four upright side supports on each side of the body. Body support longitudinal shall be not less than 10 gage steel. Body shall have sloping sides to prevent bridging of material. The rear discharge gate shall be adjustable.

3.8.2 Grid. A steel grid shall be provided on top of the body. Grid shall be constructed of not less than 1/4-inch steel rods spaced not less than 2-1/4 inches apart and centered on 1/4-inch by not less than 1-1/2 inch steel crossbars. The grid shall be constructed of four center hinged sections for easy handling. Hinge rods shall be 1/2-inch minimum diameter and run continuous from front to rear of the body.

3.8.3 Conveying system. An auger or chain-conveyor shall be used to convey the material from the hopper body to the rear discharge chute.

3.8.3.1 Auger. The steel auger shall have a progressive pitch flighting approximately 7-inch outside diameter (od), welded around on 4-inch od tube. Auger flighting shall be not less than 1/2 inch thick at the base and 3/8 inch thick at the outer edge. The outer periphery of the flights shall be mounted on self-aligning bearings on the outside of the bunk ends for ease in disassembly. The auger trough shall be easily replaceable and constructed of not less than 10 gage steel.

3.8.3.2 Conveyor and conveyor chain. The conveyor shall be constructed of 10 gage hi-grade carbon steel. The conveyor floor shall have a flat design of replaceable abrasion resistant steel. Conveyor shafts shall have dust-sealed, heavy-duty, self-aligning ball bearings equipped with greased fittings. The conveyor chain shall be of the self-cleaning pintle type, manufactured of formed alloy steel with heat treated hardened pins and chain strands of not less than 21,000 pounds tensile strength. The chain link strands shall be protected in a manner which will only expose the drag bar to the material. An adjustment device shall be provided to keep the chain tight.

3.8.4 Power system. The auger or conveyor and spinner shall be powered by independent cab-controlled hydraulic motors. A truck-mounted power takeoff or auxiliary gasoline engine shall be used to power a gear or vane type hydraulic pump. The hydraulic power system shall be complete including reservoir, valves, filters, and quick-disconnect couplers. Hydraulic hose shall conform to SAE J517, All operation controls shall be for cab mounting. The auxiliary gasoline engine, when furnished, shall be a four-cycle, air-cooled engine of not less than 18 horsepower, with electric start, charging system, ammeter, oil pressure gage (if required), and a protective hood. Operating controls shall be mounted in the truck cab.

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3.8.5 Spinner and spinner housing. The spinner and spinner housing shall be either detachable type or hinged type which raises and attaches to rear hopper body. The spinner shall be not less than 18 inches in diameter and the thickness of the replaceable fins (blades) shall be not less than 1/8 inch. The spinner shaft, bearings, and gears shall be fully enclosed and lubricated. The spinner shall be powered hydraulically. The spinner assembly shall have at least 6 inches of vertical adjustment. The spinner housing shall be fabricated of 12 gage steel minimum, and shall have an internal deflector for directing material to discharge gate. The spinner housing shall be equipped with at least three external deflectors.

3.8.6 Mounting.

3.8.6.1 Type I. Type I spreader shall be mounted on a 4 by 2, 28,000 minimum gross vehicle weight, 84-inch cab-to-axle dimension, diesel-engine-driven truck chassis conforming to KKK-T-2108, class B. Mounting of the spreader on the truck chassis shall be manufacturer's standard practice as furnished to the commercial market.

3.8.6.2 Type II. Type II spreader shall be designed for mounting in a standard 9 to 10 foot steel dump body. The manufacturer's standard holddown devices for mounting the spreader unit in the dump body shall be provided.

3.9 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator’s manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

* 3.10 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be No. 13538 yellow, conforming to FED-STD-595. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer’s current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paing shall be free from runs, sags, orange peel, or other defects.

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3.11 Lifting and tiedown attachments. When specified (see 6.2), the spreader shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the spreader. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown and lifting attachments shall be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the spreader On the carrier when. shipped.

3.12 Identification plate. An identification plate will be furnished by the contracting officer for each spreader. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a place to each spreader in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.13 Instruction plates. The spreader shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers and lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.14.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests} as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. The first article inspection shall be performed on one spreader when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. When a First Article Test is specified by this contract and the contractor desires to deliver the test unit as a contract

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item, it shall be delivered as the last item on the contract only after the contractor, at his own cost and expense, shall have completely cleaned, reconditioned and overhauled the unit, making such replacements and modifications thereto as are required to make the unit acceptable as a contract item.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, and the packaging inspection of 4.5.

4.3 Examination. Each spreader shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. When first article is required, the first article shall be tested in accordance with 4.4.1 through 4.4.4. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all spreaders until corrective action has been taken.

4.4.1 Capacity. Capacity of the spreader hopper body shall be determined either by calculation or filling with a known volume of material to determine conformance to 3.8.1. Body measurements for inside width and length shall be taken to verify dimensions specified in 3.8.1.

4.4.2 Lifting and tying down attachment test. When lifting and tying down attachments are required, they shall be tested to determine conformance to 3.11.

4.4.3 Operation test. The spreader shall be operated a sufficient period of time to determine that the equipment is in good operating condition and the spreader meets the requirements of this specification.

4.4.4 Performance test. The spreader shall be mounted as required for type I or type II. With the truck being driven on the highway, the spreader shall be capable of spreading salt or abrasive material evenly over a highway 40 feet in width, and shall be so constructed that it is easily adjustable to spreading material in an evenly distributed pattern from 6 feet to 40 feet in width.

* 4.5 Packaging inspection. The preservation, packing and marking of the spreader shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing.

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5.1.1 Boxed/crated spreaders. When specified (see 6.2), detached spreaders shall be packed in a shipping container in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. The container shall conform to the applicable carrier rules and regulations. Otherwise, the detached spreader shall be packed in such a manner as to assure safe delivery to its destination.

5.1.2 Truck-mounted spreaders. When mounted to truck chassis or body, spreaders shall be prepared for shipment in a manner that will insure adequate protection against deterioration and damage from the supplier to the first receiving activity.

* 6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The spreader covered by this specification is intended for spreading salt, sand, aggregates, or other materials on highways, streets, runways or construction sites.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of spreader required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1 and 6.4).
d. When lifting and tying down attachments are required (see 3.11).
e. When boxing/crating is required (see 5.1.1).

* 6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S29208B(YD)	920117

3. DOCUMENT TITLE
SPREADER, MATERIAL, POWER DRIVEN, TRUCK MOUNTED, COMMERCIAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
BERNADETTE E. LEWIS	805-982-5464	551-5464
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL CONSTRUCTION BATTALION CENTER (1564L) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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